



CIMA STRATEGIC CASE STUDY

Mock Exam 2

Rotomyne

Unseen

Task 1 - Unseen case material

Sodium-ion batteries, which are more suitable for efficient storage of surplus energy, especially energy from renewable sources like wind and sun, are predicted to be commercially available from the year 2030. The Board is considering investing in R&D and manufacture of sodium-ion batteries.

Around midnight on 17th November 2024 and in the early hours of the morning on 18th November 2024, Rotomyne was subject to security attacks. On the morning of 18th November 2024, you receive the following mail.

From: Martin Jacobs, Chief Finance Officer (CFO)

To: Senior Manager Finance

Subject: Cloud technology, cyber threats and penetration testing

Hi,

With the proposal to diversify into research and manufacture of sodium-ion batteries on the table, and taking into account the need to expand the capacities of our IT infrastructure, the Board feels it is time we consider investing in cloud technology. We currently have a physical data centre 20km from our Porrland Office in Capital City. Our remote backup site is located in Snowcity in the Northern part of Porrland.

Also, as you will have heard, we were victims of a physical and system attack yesterday. I have attached details of the two attacks. We do not yet know whether the two incidents are interlinked.

I need your advice on two issues before the Board meets to discuss this matter further:

- Firstly, evaluate the benefits and drawbacks of cloud technology for our organisation.

[Sub task (a) = 40%]

- Secondly, evaluate the possible consequences of these attacks and explain how penetration testing could help us prevent such attempts in future.

[Sub task (b) = 60%]

Regards,

Martin

Reference material

Details of security attacks

Time of incident: Sunday, 17th November 2024, 23:45

The SIEM (Security Information/incident and Event Management) system flagged a minor irregularity while Rotomyne's IT systems and networks underwent routine scanning. On deeper analysis, it was discovered that the integrity of the interface that enables the ICS of Yeeland factory to connect with our IT networks which connect to the WMS of our customers was compromised. Though the manufacturing operations were not disrupted it appears that an attempt was made to access confidential data.

Time of incident: Monday, 18th November, 2024 03:15

Two intruders were picked up by police who routinely patrol the outside areas of our factory in Yeeland. The intruders appear to have made an attempt to open an outside junction box that houses cables that carry data on Rotomyne's network. They attempted to run away on seeing the patrol car and were apprehended by the police.

One of the intruders had lock-picking tools that could have been used to operate the lock on the hatch that protects the cables. They also had a laptop with them, which had been confiscated by the police as evidence.

The CCTV footage from last night has been analysed, and there is no evidence of unauthorised entry into the factory itself. It appears that the intruders had the necessary skill and knowledge to avoid the security systems that protect our Yeeland factory.

Task 2 - Unseen case material

Later the same day, Martin Jacobs stops by your workplace.

“The Board is evaluating alternative scenarios regarding the response to the data threat. Here’s a print out of an email from Shaista Shameem, Chief Executive Officer, regarding the anonymous threat received.

In addition to the diversification into the manufacture of sodium-ion batteries, we need to make significant investments in strengthening our IT infrastructure. Considering the current economic conditions and the state of Porrland’s stock market, the Board has decided to fund the proposal to diversify into the manufacture of sodium-ion batteries via debt and is currently in talks with a consortium of lenders.

If the loan is approved, it would almost double our existing debt. Shaista has suggested reducing our dividend payment or not paying dividends for the current financial year. I suggested that we could look into issuing scrip dividends instead.

I need your input on two issues:

- Firstly, explain and evaluate the likely outcome if the ransom is paid but the ransomware is not triggered.

[Sub task a - 40%]

- Secondly, discuss the implications of change in dividend policy and evaluate the benefits and drawbacks to Rotomyne of issuing a scrip dividend.”

[Sub task b - 60%]

Reference material

Extracts from mail circulated by CEO to all directors

From: Shaista Shameem, Chief Executive Officer (CEO)

To: Martin Jacobs, Chief Finance Officer (CFO)

Subject: Ransom

I've just received an anonymous email threatening to encrypt our data files unless we pay P\$100 million in cryptocurrency as ransom.

We frequently receive such threats against our system, and we ensure they are evaluated carefully to determine their credibility. In this case, the email contains technical details about our hardware and software systems, suggesting a credible threat linked to the recent system attack.

If our files are encrypted, decryption is nearly impossible. Furthermore, the attacker could copy sensitive information, including Personally Identifiable Financial Information (PIFI) and details about ongoing research projects, before encryption occurs. The potential exposure of this data poses significant risks to our reputation and operational integrity.

While we maintain a hot backup site, the backup files are also at risk of encryption during an attack. This redundancy does not guarantee protection if the ransomware is executed.

The email stipulates a 48-hour window to pay the ransom before the ransomware is triggered. We have been explicitly warned not to contact the police, which adds to the urgency and complexity of our response strategy.

In light of these circumstances, we must consider our options carefully. Engaging cybersecurity experts to assess the situation and implement preventative measures is crucial, as is preparing for potential legal and reputational ramifications should we face a data breach.

Task 3 - Unseen case material

The next morning you receive the following email.

From: Martin Jacobs, Chief Finance Officer (CFO)

To: Senior Finance Manager

Subject: Internal Controls and Corporate Governance

Hi,

We engaged with an external cybersecurity consultant on a high-priority basis who worked through the night yesterday and successfully removed the malware from our system and protected our data files. Meanwhile, we undertook our own internal investigations, details of which are attached.

In light of these cybersecurity incidents, Pierre Forcier, Senior Independent Director, has suggested that Rotomyne should appoint an additional non-executive director with a background in cybersecurity.

I need your advice on two matters:

- Firstly, recommend and justify four internal controls (other than penetration testing) that we might introduce to prevent the recurrence of this type of threat.

[Sub task a - 50%]

- Secondly, evaluate the claim that we should appoint an additional non-executive director with a background in cybersecurity.

[Sub task b - 50%]

Regards,

Martin

Reference material

Extract from Preliminary Investigation Report

The Security Information and Event Management (SIEM) system detected a minor irregularity during routine scanning of Rotomyne's IT systems and networks, though the full impact of the event was initially unclear.

It appears that a Senior Inventory Manager at our Yeeland factory, which supplies more than 60% of our products to our largest customer, the vehicle manufacturer Gord, clicked on a fraudulent email that appeared to come from Gord's head office. This action led to a Structured Query Language (SQL) injection being introduced into our system.

At this stage, it is uncertain whether the staff member was tricked or bribed into clicking the email.

SQL is a standard computer language used to store, retrieve, and manage data in databases. In an SQL injection attack, the cyber attacker inserts malicious SQL code into a database to exploit its vulnerabilities. This malicious code can retrieve confidential information by tricking the system into releasing sensitive data or, in some cases, can be used to destroy the data entirely.

Present value tables

Present value of US\$1 i.e. $(1 + r)^{-n}$ where r = interest rate, n = number of periods until payment or receipt.

Period s (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	.990	.980	.971	.962	.962	.943	.935	.926	.917	.909
2	.980	.961	.943	.925	.907	.890	.873	.857	.842	.826
3	.971	.942	.915	.889	.864	.840	.816	.794	.772	.751
4	.961	.924	.888	.855	.823	.792	.763	.735	.708	.683
5	.951	.906	.863	.822	.784	.747	.713	.681	.650	.621
6	.942	.888	.837	.790	.746	.705	.666	.630	.596	.564
7	.933	.871	.813	.760	.711	.665	.623	.583	.547	.513
8	.923	.853	.789	.731	.677	.627	.582	.540	.502	.467
9	.914	.837	.766	.703	.645	.592	.544	.500	.460	.424
10	.905	.820	.744	.676	.614	.558	.508	.463	.422	.386
11	.896	.804	.722	.650	.585	.527	.475	.429	.388	.350
12	.887	.788	.701	.625	.557	.497	.444	.397	.356	.319
13	.879	.773	.681	.601	.530	.469	.415	.368	.326	.290
14	.870	.758	.661	.577	.505	.442	.388	.340	.299	.263
15	.861	.743	.642	.555	.481	.417	.362	.315	.275	.239
16	.853	.728	.623	.534	.458	.394	.339	.292	.252	.218
17	.844	.714	.605	.513	.436	.371	.317	.270	.231	.198
18	.836	.700	.587	.494	.416	.350	.296	.250	.212	.180
19	.828	.686	.570	.475	.396	.331	.277	.232	.194	.164
20	.820	.673	.554	.456	.377	.312	.258	.215	.178	.149

Period s (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	.901	.893	.885	.877	.870	.862	.855	.847	.840	.833
2	.812	.797	.783	.769	.756	.743	.731	.718	.706	.694
3	.731	.712	.693	.675	.658	.641	.624	.609	.593	.579
4	.659	.636	.613	.592	.572	.552	.534	.516	.499	.482
5	.593	.567	.543	.519	.497	.476	.456	.437	.419	.402
6	.535	.507	.480	.456	.432	.410	.390	.370	.352	.335
7	.482	.452	.425	.400	.376	.354	.333	.314	.296	.279
8	.434	.404	.376	.351	.327	.305	.285	.266	.249	.233
9	.391	.361	.333	.308	.284	.263	.243	.225	.209	.194
10	.352	.322	.295	.270	.247	.227	.208	.191	.176	.162
11	.317	.287	.261	.237	.215	.195	.178	.162	.148	.135
12	.286	.257	.231	.208	.187	.168	.152	.137	.124	.112
13	.258	.229	.204	.182	.163	.145	.130	.116	.104	.093
14	.232	.205	.181	.160	.141	.125	.111	.099	.088	.078
15	.209	.183	.160	.140	.123	.108	.095	.084	.074	.065
16	.188	.163	.141	.123	.107	.093	.081	.071	.062	.054
17	.170	.146	.125	.108	.093	.080	.069	.060	.052	.045
18	.153	.130	.111	.095	.081	.069	.059	.051	.044	.038
19	.138	.116	.098	.083	.070	.060	.051	.043	.037	.031
20	.124	.104	.087	.073	.061	.051	.043	.037	.031	.026

Cumulative present value tables

This table shows the Present Value of US\$1 per annum, Receivable or Payable at the end of each year for n years

$$\frac{1 - (1+r)^{-n}}{r}$$

Periods (n)	Interest rates (r)									
	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606
16	14.718	13.578	12.561	11.652	10.838	10.106	9.447	8.851	8.313	7.824
17	15.562	14.292	13.166	12.166	11.274	10.477	9.763	9.122	8.544	8.022
18	16.398	14.992	13.754	12.659	11.690	10.828	10.059	9.372	8.756	8.201
19	17.226	15.679	14.324	13.134	12.085	11.158	10.336	9.604	8.950	8.365
20	18.046	16.351	14.878	13.590	12.462	11.470	10.594	9.818	9.129	8.514
Periods (n)	Interest rates (r)									
	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12	6.492	6.194	5.918	5.660	5.421	5.197	4.968	4.793	4.611	4.439
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675
16	7.379	6.974	6.604	6.265	5.954	5.668	5.405	5.162	4.938	4.730
17	7.549	7.120	6.729	6.373	6.047	5.749	5.475	5.222	4.990	4.775
18	7.702	7.250	6.840	6.467	6.128	5.818	5.534	5.273	5.033	4.812
19	7.839	7.366	6.938	6.550	6.198	5.877	5.584	5.316	5.070	4.843
20	7.963	7.469	7.025	6.623	6.259	5.929	5.628	5.353	5.101	4.870